Series

Chapter Overview

Method of differences

Split a summand into terms which cancel.

Write the first and last terms explicitly.

Retain every uncancelled boundary term.

Maclaurin series

Match derivatives at zero.

Use standard expansions and substitutions.

State the interval of convergence when required.

STARTER – Round 1!

Solution 3

1. Find the product

$$\frac{1}{2} \times \frac{2}{3} \times \cdots \times \frac{n-1}{n}$$

2. Find

$$\prod_{r=2}^{n} \left(1 - \frac{1}{r^2}\right)$$

1

$$\frac{1 \cdot 2 \cdots (n-1)}{2 \cdot 3 \cdots n} = \frac{1}{n}$$

2

$$1 - \frac{1}{r^2} = \frac{(r-1)(r+1)}{r^2}$$

$$\prod_{r=2}^{n} \frac{r-1}{r} \prod_{r=2}^{n} \frac{r+1}{r}$$

$$= \frac{1}{n} \frac{n+1}{2} = \frac{n+1}{2n}$$

STARTER – Round 2!

Solution 4

Use cancellation to find

$$S_n = \sum_{r=1}^{n} \left(\frac{1}{r} - \frac{1}{r+1}\right)$$

Sum

$$S_n = (1 - \tfrac{1}{2}) + (\tfrac{1}{2} - \tfrac{1}{3}) + \cdots + (\tfrac{1}{n} - \tfrac{1}{n+1})$$

$$= 1 - \frac{1}{n+1} = \frac{n}{n+1}$$

Example

Solution 5

Use the identity

$4r^3 = r^2(r+1)^2 - (r-1)^2r^2$

to find $\sum_{r=1}^{n} r^3$.

Sum

$$4\sum_{r=1}^{n} r^3 = (1^2 2^2 - 0) + (2^2 3^2 - 1^2 2^2) + \cdots$$

$$4\sum_{r=1}^{n} r^3 = +n^2(n+1)^2 - (n-1)^2 n^2$$

$$\sum_{r=1}^{n} r^3 = \frac{n^2(n+1)^2}{4}$$

Partial Fraction Example

Solution 6

Find

$$\sum_{r=1}^{n} \frac{1}{4r^2 - 1}$$

Partial fractions

$$\frac{1}{4r^2-1} = \frac{1}{2}\left(\frac{1}{2r-1} - \frac{1}{2r+1}\right)$$

Sum

$$S_n = \frac{1}{2}\left(1 - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \cdots + \frac{1}{2n-1} - \frac{1}{2n+1}\right)$$

$$= \frac{1}{2}\left(1 - \frac{1}{2n+1}\right) = \frac{n}{2n+1}$$

Test Your Understanding

Solution 7

a. Express $2/((2r+1)(2r+3))$ in partial fractions.

b. Hence sum $3/((2r+1)(2r+3))$, for $r = 1, \ldots, n$.

a

$$\frac{2}{(2r+1)(2r+3)} = \frac{1}{2r+1} - \frac{1}{2r+3}$$

b

$$S_n = \frac{3}{2}\left(\frac{1}{3} - \frac{1}{2n+3}\right) = \frac{n}{2n+3}$$

Harder telescoping sums

Solution 8

Find

$$\sum_{r=1}^{n}\left(\frac{1}{r} - \frac{1}{r+2}\right)$$

Sum

$$S_n = 1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}$$

$$= \frac{n(3n+5)}{2(n+1)(n+2)}$$

Two terms survive at each end.

Test Your Understanding — Solution 9

1a. Express $2/((r+1)(r+3))$ in partial fractions.

b. Sum from $r = 1$ to n.

c. Evaluate the sum from $r = 10$ to 100.

a

$$\frac{2}{(r+1)(r+3)} = \frac{1}{r+1} - \frac{1}{r+3}$$

b

$$S_n = \frac{1}{2} + \frac{1}{3} - \frac{1}{n+2} - \frac{1}{n+3}$$
$$= \frac{n(5n+13)}{6(n+2)(n+3)}$$

c

$$S_{100} - S_9 = \frac{1}{11} + \frac{1}{12} - \frac{1}{102} - \frac{1}{103}$$
$$= \frac{11921}{77044} \approx 0.1547$$

Exercise 2A - Core Pure Year 2 Pages 36-37 — Solution 10

2. Show that

$$\frac{1}{r(r+1)(r+2)} = \frac{1}{2r(r+1)} - \frac{1}{2(r+1)(r+2)}$$

and hence sum the left side from 1 to n.

Identity

The right side has numerator $(r+2) - r = 2$ over $2r(r+1)(r+2)$.

Sum

$$S_n = \frac{1}{4} - \frac{1}{2(n+1)(n+2)}$$
$$= \frac{n(n+3)}{4(n+1)(n+2)}$$

3. Use partial fractions and the method of differences to sum

$$\sum_{r=1}^{n} \frac{1}{r(r+2)}$$

3

$$\frac{1}{r(r+2)} = \frac{1}{2}\left(\frac{1}{r} - \frac{1}{r+2}\right)$$

$$S_n = \frac{1}{2}\left(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}\right)$$

$$= \frac{n(3n+5)}{4(n+1)(n+2)}$$

4. Use partial fractions to find

$$\sum_{r=1}^{n} \frac{1}{(r+2)(r+3)}$$

4

$$\frac{1}{(r+2)(r+3)} = \frac{1}{r+2} - \frac{1}{r+3}$$

$$S_n = \frac{1}{3} - \frac{1}{n+3} = \frac{n}{3(n+3)}$$

5. Prove that
$r/(r+1)! = 1/r! - 1/(r+1)!$.
Hence find
$$\sum_{r=1}^{n} \frac{r}{(r+1)!}$$

5
$$\frac{1}{r!} - \frac{1}{(r+1)!} = \frac{r+1-1}{(r+1)!}$$
$$S_n = 1 - \frac{1}{(n+1)!}$$

6. Show that
$$\frac{2r+1}{r^2(r+1)^2} = \frac{1}{r^2} - \frac{1}{(r+1)^2}$$
and hence sum from 1 to n.

6
The numerator of the difference is
$(r+1)^2 - r^2 = 2r + 1$.
$$S_n = 1 - \frac{1}{(n+1)^2} = \frac{n(n+2)}{(n+1)^2}$$

8. Find constants a, b such that

$$\sum_{r=1}^{n} \frac{8}{(3r-2)(3r+4)} = \frac{n(an+b)}{(3n+1)(3n+4)}$$

8

$$\frac{8}{(3r-2)(3r+4)} = \frac{4}{3}\left(\frac{1}{3r-2} - \frac{1}{3r+4}\right)$$

$$S_n = \frac{4}{3}\left(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4}\right)$$

$$= \frac{n(15n+17)}{(3n+1)(3n+4)}$$

$$a = 15, \quad b = 17$$

9. Find a if

$$\sum_{r=1}^{n}((r+1)^2 - (r-1)^2) = an(n+1)$$

9

$$\sum 4r = 2n(n+1) \Rightarrow a = 2$$

10a. Write $\sum_{r=1}^{n} 3/((3r+1)(3r+4))$ as $an/(bn+c)$.

b. Sum the same expression from $r = n$ to $2n$.

10a

$$S_n = \frac{1}{4} - \frac{1}{3n+4} = \frac{3n}{12n+16}$$

$$a = 3, \; b = 12, \; c = 16$$

10b

$$S_{2n} - S_{n-1} = \frac{1}{3n+1} - \frac{1}{6n+4}$$

$$= \frac{3(n+1)}{2(3n+1)(3n+2)}$$

11. Robin claims

$$\sum_{r=1}^{n} \frac{2r+1}{r(r+1)} = 1 - \frac{1}{n+1}$$

Explain the error in using the method of differences here.

11

$$\frac{2r+1}{r(r+1)} = \frac{1}{r} + \frac{1}{r+1}$$

The partial fractions are added, not subtracted. Successive terms do not cancel.

For example, when $n = 1$, the sum is $3/2$, whereas the claimed expression is $1/2$.

12. Find a, b in

$$\sum_{r=1}^{n} \frac{1}{r(r+2)} = \frac{3}{4} - \frac{an+b}{2(n+1)(n+2)}$$

12

$$S_n = \frac{3}{4} - \frac{1}{2}\left(\frac{1}{n+1} + \frac{1}{n+2}\right)$$

$$= \frac{3}{4} - \frac{2n+3}{2(n+1)(n+2)}$$

$$a = 2, \quad b = 3$$

13a Express $4/((2r+1)(2r+5))$ in partial fractions.

b. Evaluate the sum from $r = 16$ to 25, to four decimal places.

a

$$\frac{4}{(2r+1)(2r+5)} = \frac{1}{2r+1} - \frac{1}{2r+5}$$

b

$$\sum_{r=16}^{25} \frac{4}{(2r+1)(2r+5)} = \frac{1}{33} + \frac{1}{35} - \frac{1}{53} - \frac{1}{55}$$

$$\approx 0.0218$$

Higher derivatives

Differentiate repeatedly. The second derivative is the derivative of the first derivative.

$f(x), \quad f'(x), \quad f''(x), \quad f^{(n)}(x)$

$\frac{dy}{dx}, \quad \frac{d^2y}{dx^2}, \quad \frac{d^ny}{dx^n}$

Newton notation $\dot{y}, \ddot{y}$ is commonly used for derivatives with respect to time.

Examples

Solution 21

1. Find the third derivative of $\ln(1-x)$ at $x = 1/2$.
2. For $f = e^{x^2}$, find the first three derivatives at zero.

1

$f' = -(1-x)^{-1}$

$f'' = -(1-x)^{-2}$

$f''' = -2(1-x)^{-3} \Rightarrow f'''(1/2) = -16$

2

$f' = 2xf$

$f'' = 2f + 2xf'$

$f''' = 4f' + 2xf''$

$f'(0) = 0,\ f''(0) = 2,\ f'''(0) = 0$

From derivatives to polynomials

Solution 22

From derivatives to polynomials

Can a polynomial reproduce the value and derivatives of a function at a chosen point?

At $x = 0$, the coefficients of $a + bx + cx^2 + dx^3 + \cdots$ determine its successive derivatives.

$f(0) = a, \quad f'(0) = b$

$f''(0) = 2c, \quad f'''(0) = 6d$

Why we might represent functions as power series

Solution 23

Why use power series?

Polynomials are easy to evaluate, differentiate and integrate.

A series can approximate functions near a chosen point.

Applications include numerical integration, approximating constants and solving differential equations.

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$$

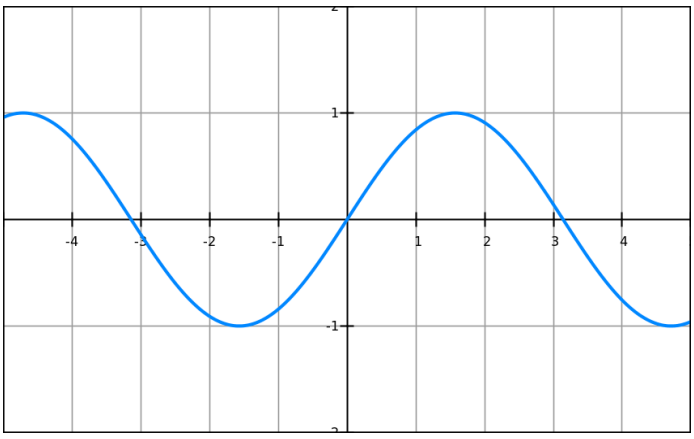
At $x = 1$, this gives a series for $\pi/4$.

Now onto the good stuff!

Solution 24

Match the value of $\sin x$ at zero with a polynomial.

$f(x) \approx a + bx + cx^2 + dx^3$



Coefficient

$f(0) = 0 \Rightarrow a = 0$

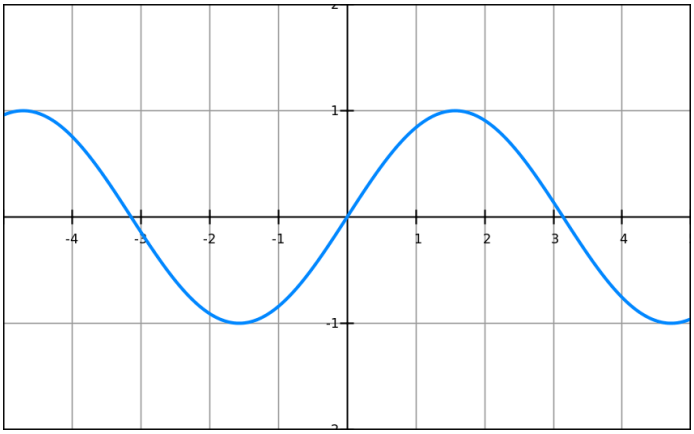
Match successive derivatives at $x = 0$

Maclaurin Expansion

Solution 25

Now match its first derivative at zero.

$f(x) \approx a + bx + cx^2 + dx^3$



Coefficient

$f'(0) = \cos 0 = 1 \Rightarrow b = 1$

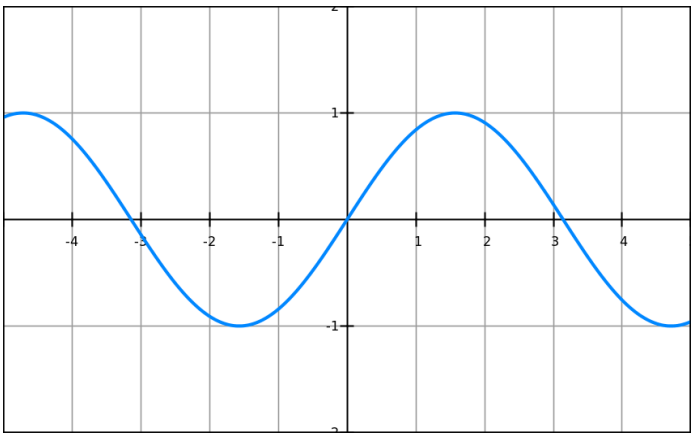
Match successive derivatives at $x = 0$

Maclaurin Expansion

Solution 26

Now match its second derivative at zero.

$f(x) \approx a + bx + cx^2 + dx^3$



Coefficient

$f''(0) = -\sin 0 = 0 \Rightarrow c = 0$

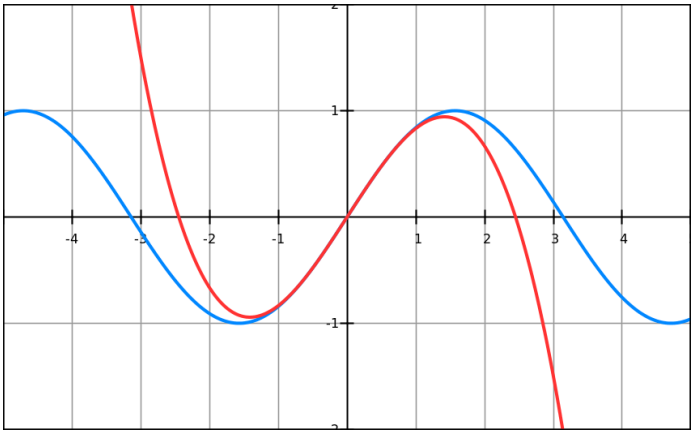
Match successive derivatives at $x = 0$

Maclaurin Expansion

Solution 27

Match the third derivative at zero.

$f(x) \approx a + bx + cx^2 + dx^3$



Coefficient

$f'''(0) = -\cos 0 = -1 \Rightarrow d = -1/6$

$\sin x \approx x - \dfrac{x^3}{6}$

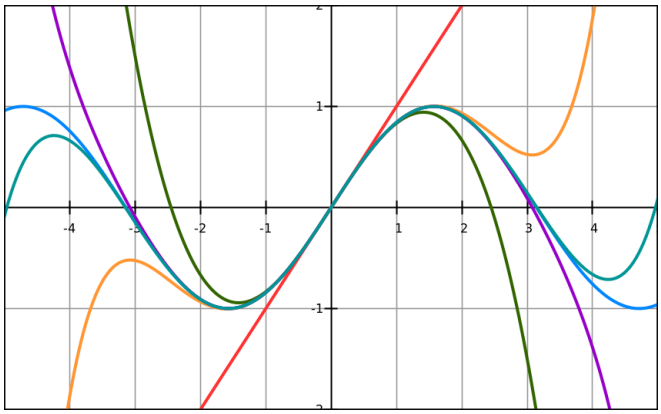
Maclaurin Expansion

Solution 28

Maclaurin expansion

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots$$

Equality with an infinite series holds only where that series converges to the function. A truncated series gives an approximation.



$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

Example

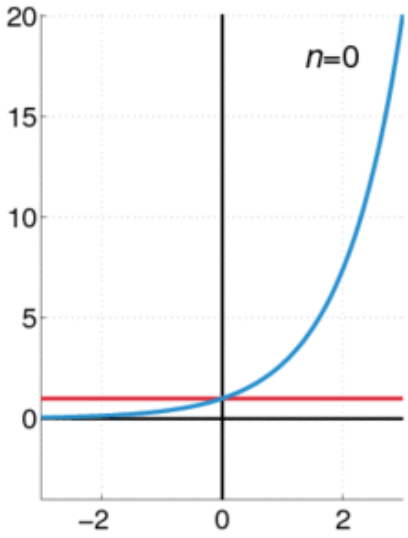
Solution 29

Standard series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Both converge for every real x.



The animation shows polynomial approximations improving near the expansion point.

Example

Solution 30

Use the first three non-zero terms to estimate $\ln 1.05, \ln 1.25, \ln 1.8$.

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

Estimates

$\ln 1.05 \approx 0.0487917$

$\ln 1.25 \approx 0.223958$

$\ln 1.8 \approx 0.650667$

The approximation is more accurate when x is closer to zero.

Example

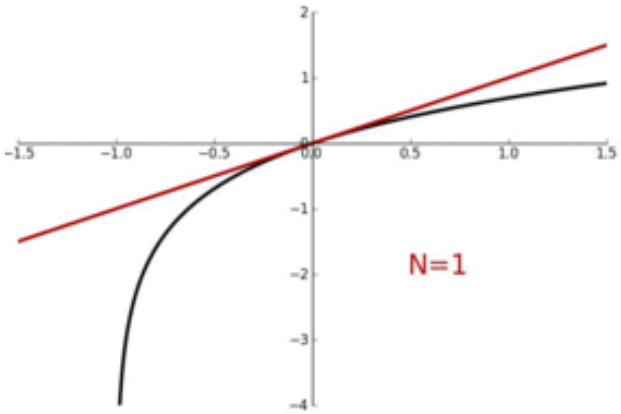
Solution 31

Convergence matters

$$\ln(1+x) = \sum_{r=1}^{\infty} \frac{(-1)^{r+1}x^r}{r}$$

$-1 < x \leq 1$

At $x = 1$, the alternating harmonic series converges to $\ln 2$. At $x = -1$, it diverges.



Exercise 2C - Core Pure Year 2 Pages 43-44

Solution 32

1a. Obtain the Maclaurin series for $(1-x)^{-1}$.

b. Find the first four terms for $\sqrt{1+x}$.

a

$$f^{(r)}(x) = r!(1-x)^{-(r+1)}$$

$$f^{(r)}(0) = r!$$

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots, \quad |x| < 1$$

b

$$f(0) = 1, \; f'(0) = \frac{1}{2}$$

$$f''(0) = -\frac{1}{4}, \; f'''(0) = \frac{3}{8}$$

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} + \cdots$$

2. Find the first three terms of the Maclaurin expansion of $e^{\sin x}$ by differentiation.

2

$f' = \cos x\, e^{\sin x}$

$f'' = (\cos^2 x - \sin x)e^{\sin x}$

$f(0) = f'(0) = f''(0) = 1$

$e^{\sin x} = 1 + x + \frac{x^2}{2} + \cdots$

3a. Derive the Maclaurin series for $\cos x$.

b. Use three non-zero terms to estimate $\cos 30°$.

a

The derivatives at zero cycle through $1, 0, -1, 0$.

$\cos x = \sum_{r=0}^{\infty} \frac{(-1)^r x^{2r}}{(2r)!}$

b

Convert to radians: $x = \pi/6$.

$\cos 30° \approx 1 - \frac{(\pi/6)^2}{2} + \frac{(\pi/6)^4}{24}$

$\approx 0.866053 \approx 0.866 \quad (3 \text{ d.p.})$

6. Find the first five terms in the Maclaurin expansion of $\cos(x - \pi/4)$.

6

$$\cos(x - \pi/4) = \frac{1}{\sqrt{2}}(\cos x + \sin x)$$
$$= \frac{1}{\sqrt{2}}\left(1 + x - \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} + \cdots\right)$$

7. Let $f(x) = (1 - x)^2 \ln(1 - x)$.
a. Find f' and f''.
b. Evaluate the first three derivatives at zero.
c. Give the first three non-zero Maclaurin terms.

a

$$f' = x - 1 - 2(1 - x)\ln(1 - x)$$
$$f'' = 3 + 2\ln(1 - x)$$

b

$$f(0) = 0,\ f'(0) = -1$$
$$f''(0) = 3,\ f'''(0) = -2$$

c

$$f(x) = -x + \frac{3}{2}x^2 - \frac{1}{3}x^3 + \cdots$$

8. Find the first two non-zero terms of $3\sin x - 4x\cos x + x$.

Hence find its ratio to x^3 as $x \to 0$.

a

$$3(x - x^3/6 + x^5/120) - 4x(1 - x^2/2 + x^4/24) + x$$
$$= \frac{3}{2}x^3 - \frac{17}{120}x^5 + \cdots$$

b

$$\lim_{x\to 0}\left(\frac{3}{2} - \frac{17}{120}x^2 + \cdots\right) = \frac{3}{2}$$

9. Let $f(x) = \ln(\cos x)$.
   a. Show $f'(x) = -\tan x$.
   b. Find the derivatives at zero up to order four.
   c. Give its first two non-zero Maclaurin terms.

a

$$f' = \frac{-\sin x}{\cos x} = -\tan x$$

b

$$f(0) = 0,\ f'(0) = 0,\ f''(0) = -1$$
$$f'''(0) = 0,\ f^{(4)}(0) = -2$$

c

$$\ln(\cos x) = -\frac{x^2}{2} - \frac{x^4}{12} + \cdots$$

10. Find the Maclaurin expansion of $\tan x$, up to and including x^5.

10

Use $f' = 1 + f^2$ and differentiate repeatedly. At zero:

$f = 0,\ f' = 1,\ f'' = 0$

$f''' = 2,\ f^{(4)} = 0,\ f^{(5)} = 16$

$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$

Composite Functions

Substituting into standard series

$$e^x = \sum_{r=0}^{\infty} \frac{x^r}{r!}$$

$$\sin x = x - \frac{x^3}{3!} + \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \cdots$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$$

Find the first four non-zero terms of $\cos(2x^2)$.

Expansion

$\cos(2x^2) = 1 - \frac{(2x^2)^2}{2!} + \frac{(2x^2)^4}{4!} - \frac{(2x^2)^6}{6!} + \cdots$

$= 1 - 2x^4 + \frac{2}{3}x^8 - \frac{4}{45}x^{12} + \cdots$

Composite Functions — Solution 41

Expand to x^3 and state the interval of validity:

$$\ln \frac{\sqrt{1+2x}}{1-3x}$$

Expansion

$$\frac{1}{2}\ln(1+2x) - \ln(1-3x)$$
$$= \frac{1}{2}(2x - 2x^2 + \tfrac{8}{3}x^3) + (3x + \tfrac{9}{2}x^2 + 9x^3) + \cdots$$
$$= 4x + \frac{7}{2}x^2 + \frac{31}{3}x^3 + \cdots$$

Validity

$$-\frac{1}{3} \le x < \frac{1}{3}$$

Composite Functions — Solution 42

Expand $e^{\sin x}$ through the term in x^4, using standard series.

Expansion

$$e^{\sin x} = 1 + \sin x + \frac{\sin^2 x}{2} + \frac{\sin^3 x}{6} + \frac{\sin^4 x}{24} + \cdots$$
$$= 1 + (x - x^3/6) + \tfrac{1}{2}(x^2 - x^4/3) + x^3/6 + x^4/24 + \cdots$$
$$= 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + O(x^5)$$

Exercise 2D - Core Pure Year 2 Pages 46-48 — Solution 43

2a. Expand $\ln((1+x)/(1-x))$.

b. Hence expand the logarithm of its square root.

c. Estimate $\ln(2/3)$, using three terms.

d. Estimate $\ln 2$, and compare to the calculator value.

a

$$\ln\frac{1+x}{1-x} = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right)$$

Valid for $-1 < x < 1$.

b

$$\ln\sqrt{\frac{1+x}{1-x}} = x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots$$

c

$$x = -\frac{1}{5}: \quad \ln(2/3) \approx -0.4055$$

d

For the square-root expression, take $x = 3/5$.

$$\ln 2 \approx 0.6 + \frac{0.6^3}{3} + \frac{0.6^5}{5} = 0.687552$$

Both this and the calculator value $0.693147\ldots$ round to 0.69 (2 d.p.).

Exercise 2D - Core Pure Year 2 Pages 46-48 — Solution 44

3. Show that, for small x, $e^{2x} - e^{-x} \approx 3x + 3x^2/2$.

3

$$(1 + 2x + 2x^2) - (1 - x + x^2/2) = 3x + \frac{3}{2}x^2$$

4a. Expand $3x\sin 2x - \cos 3x$ through x^4.

b. Find the limit of $(3x\sin 2x - \cos 3x + 1)/x^2$ as $x \to 0$.

4a

$$3x(2x - \tfrac{4}{3}x^3) - (1 - \tfrac{9}{2}x^2 + \tfrac{27}{8}x^4)$$
$$= -1 + \frac{21}{2}x^2 - \frac{59}{8}x^4 + \cdots$$

4b

$$\text{Limit} = \frac{21}{2}$$

6a. Expand $\cos 2x$ through x^8.

b. Hence obtain the expansion of $\sin^2 x$ through x^8.

a

$$\cos 2x = 1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6 + \frac{2}{315}x^8 + \cdots$$

b

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$
$$= x^2 - \frac{x^4}{3} + \frac{2x^6}{45} - \frac{x^8}{315} + \cdots$$

7. The first non-zero terms of $\ln(1+x) + (x-1)(e^x - 1)$ are $px^3 + qx^4$. Find p, q.

7

$$(x-1)(e^x - 1) = -x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{8} + \cdots$$
$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$
$$p = \frac{2}{3}, \qquad q = -\frac{1}{8}$$

8a. Expand $\sin x/(1-x)^2$ through x^4.

b. Find the gradient of this function at the origin.

a

$$\frac{\sin x}{(1-x)^2} = (x - x^3/6 + \cdots)(1 + 2x + 3x^2 + 4x^3 + \cdots)$$

$$= x + 2x^2 + \frac{17}{6}x^3 + \frac{11}{3}x^4 + \cdots$$

b

The coefficient of x is 1, so the gradient at zero is 1.

10a. Give the first five non-zero terms of $e^{-x^2/2}$.

b. Integrate this approximation to estimate $\int_{-1}^{1} e^{-x^2/2}\,dx$.

a

$$e^{-x^2/2} = 1 - \frac{x^2}{2} + \frac{x^4}{8} - \frac{x^6}{48} + \frac{x^8}{384} + \cdots$$

b

$$\int_{-1}^{1} e^{-x^2/2}\,dx \approx 2\left(1 - \frac{1}{6} + \frac{1}{40} - \frac{1}{336} + \frac{1}{3456}\right)$$

$$\approx 1.711 \quad (3 \text{ d.p.})$$

11a. Expand $e^{px}\sin 3x$ through x^3.

b. If $e^{px}\sin 3x + \ln(1+qx) - x$ has leading term kx^3, find p, q, k.

a

$$e^{px}\sin 3x = 3x + 3px^2 + \frac{3}{2}(p^2-3)x^3 + \cdots$$

b

$$3 + q - 1 = 0 \Rightarrow q = -2$$

$$3p - q^2/2 = 0 \Rightarrow p = 2/3$$

$$k = \frac{3}{2}(p^2-3) + \frac{q^3}{3} = -\frac{13}{2}$$

12. For $x > 0$, let $f(x) = e^{x-\ln x}\sin x$.

a. Expand, neglecting x^4 and higher powers.

b. Estimate $f(0.1)$ and compare with a calculator.

a

$$f(x) = e^x \frac{\sin x}{x}$$

$$= (1 + x + x^2/2 + x^3/6 + \cdots)(1 - x^2/6 + \cdots)$$

$$= 1 + x + \frac{x^2}{3} + O(x^4)$$

b

$f(0.1) \approx 1.103333\ldots$

Calculator: $1.103329\ldots$. Both give 1.10333 to six significant figures.

13. Let $y = \sin 2x - \cos 2x$.

a. Show $y'' = -4y$, and hence $y^{(4)} = 16y$.

b. Find its first five non-zero Maclaurin terms.

a

$y' = 2\cos 2x + 2\sin 2x$

$y'' = -4\sin 2x + 4\cos 2x = -4y$

$y^{(4)} = -4y'' = 16y$

b

$y = -1 + 2x + 2x^2 - \frac{4}{3}x^3 - \frac{2}{3}x^4 + \cdots$